Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 6 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 6 : SLO â€“ 2 | Matrix Multiplication with DMA â€” Compute and Print

I. Fill in the Missing Code

1. Input mat1: for (i = 0; i < r1; i++) for (j = 0; j < c1; j++) scanf("%d", &mat1[i][j]);
2. Input mat2: for (i = 0; i < r2; i++) for (j = 0; j < c2; j++) scanf("%d", &mat2[i][j]);
3. Multiplication: for (i = 0; i < r1; i++) for (j = 0; j < c2; j++) for (k = 0; k < c1; k++) res[i][j] += mat1[i][k] * mat2[k][j];
4. Print: for (i = 0; i < r1; i++) { for (j = 0; j < c2; j++) printf("%d ", res[i][j]); printf("\n"); }

II. Cleanup (good practice)

• for (i = 0; i < r1; i++) { free(mat1[i]); free(res[i]); } free(mat1); free(res);
• for (i = 0; i < r2; i++) free(mat2[i]); free(mat2);